

Figure 8: Accuracy under FGM attack for a) the BNN and b) the SIN 0.05 on FashionMNIST and c) ResNet with dropout probability 0.3 on CIFAR10 for different perturbation strengths and amount of samples used for calculating the attack. During inference we used 100 samples.

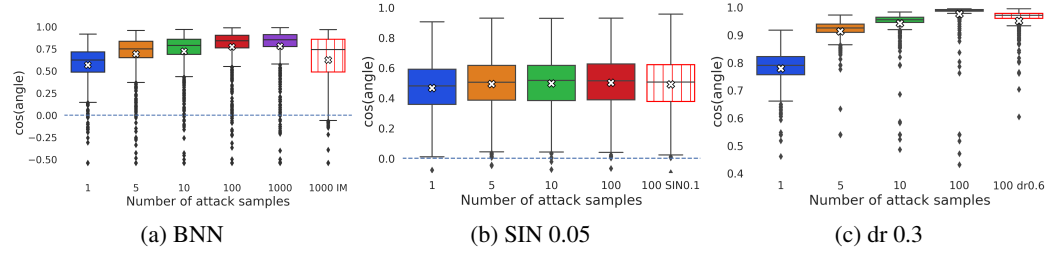


Figure 9: Cosine of the angle for the first 1,000 test set images from the FashionMNIST and CIFAR10 test set for an a) BNN, b) SIN 0.05 and c) ResNet trained with dropout probability 0.3 when attacked with different amounts of samples and attack strength 1.5 and 0.3 with FGM respectively. White crosses indicate the mean value.

C.1 Complementary experiments on accuracy of robustness conditions

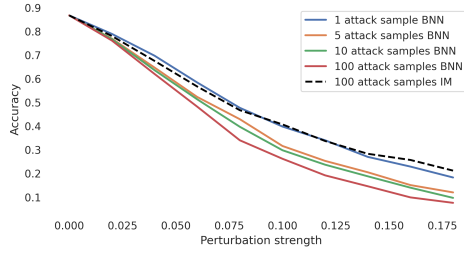
For completeness we attached the results on the transferability of our derived sufficient conditions to: the IM on FashionMNIST and the two ResNet with dropout probability 0.3 and 0.6 on CIFAR10. For the BNN we used the same setting as described in the main paper and derive similar results (c.f. figure 7): while the percentage of samples fulfilling the condition $\min_c r_c^T > \|\delta^A\|_2$ approaches zero with growing perturbation strength the percentage of samples fulfilling the condition from theorem 1 in the main paper closely matches the real adversarial accuracy in a narrow environment. For models on CIFAR10 we had to adapt the noise added for the smooth classifier to 0.01 and reduce the amount of samples during inference to 50 such that it fits on one GPU.

C.2 Complementary experiments on stronger attacks

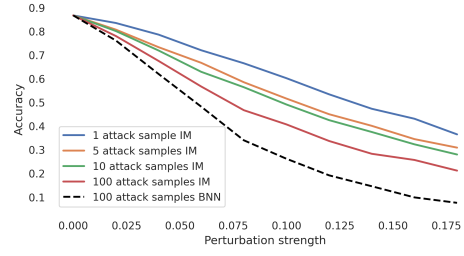
We first present the results not shown in the main paper. That is, we investigate the accuracy under FGM attack with an increasing amount of samples used during the attack (c.f. figure 8). Similar to the observations in the main paper, the accuracy under attack is reduced by an increased amount of samples. However, we observe only a very small decrease in accuracy when increasing the amount of samples from 100 and 1,000 for the BNN, from 1 to 5 or above for the SIN 0.05 and when using 5 instead of 10 or 100 samples for the attack on the ResNet trained with dropout probability 0.3. This observation is mirrored by the reduction of $\cos(\alpha_c^{T,A})$ displayed in figure 9 where we observe an increase of the cosine boxplots which matches the decrease of the adversarial accuracy when taking more samples during the attack.

C.2.1 Attacks with FGSM (ℓ_∞ - norm)

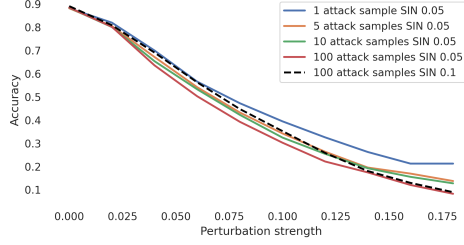
All adversarial examples in the main part of the paper were based on an ℓ_2 -norm constraint, which we chose for the nice geometric distance interpretation. However, the first proposed attack scheme [Goodfellow et al., 2015] was based on ℓ_∞ -norm, which we test in the following. In figure 10 and 11 we see the respective results on the different data sets. For all models the robustness is decreased with



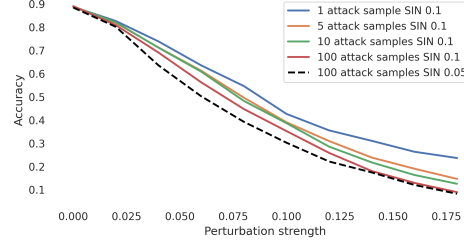
(a) BNN



(b) IM

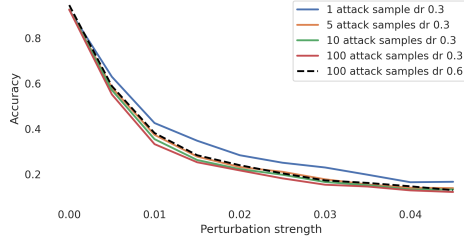


(c) SIN 0.05

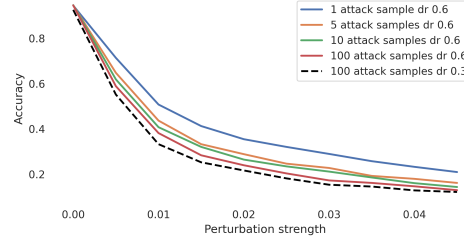


(d) SIN 0.1

Figure 10: Accuracy under FGSM attack under ℓ_∞ -norm constraint for a) the BNN, b) the IM, c) SIN 0.05 and d) SIN 0.1 on the first 1,000 test set images from FashionMNIST for different perturbation strengths and amount of samples used for calculating the attack. Predictions during inference are based on 100 samples.



(a) droprate 0.3



(b) droprate 0.6

Figure 11: Accuracy under FGSM attack under ℓ_∞ -norm constraint for the ResNet model with a dropout probability a) of 0.3 and b) of 0.6 on the first 1,000 test set images from CIFAR10 for different perturbation strengths and amount of samples used for calculating the attack. Predictions during inference are based on 100 samples.

multiple samples and models with higher prediction variance also have a higher accuracy under this attack. Note, that the values of the perturbation strength are not comparable to the values under ℓ_2 -norm constraint, since $\|x\|_\infty \leq \|x\|_2$.

C.2.2 Attacks with PGD

Projected gradient descent [Madry et al., 2018] is a strong iterative attack, where multiple small steps of size ν of fast gradient method are applied. Specifically, we used the same ℓ_2 -norm length constraint on η as in the experiments of the main part of the paper but chose step size $\nu = \eta/50$ and 100 iterations. Note that at each iteration a new network is sampled such that for an attack based on 1 samples, 100 different attack networks were seen, for an attack based on 5 samples 500 different networks and so on. In figure 12 and 13 we see that the overall accuracy is decreased compared to the results for FGM, but still, IM has a higher accuracy under attack than the BNN and so does the ResNet with a higher dropout probability. Further, the attacks get stronger with taking more samples, so the general observations made in the main paper also hold for strong attacks and are not due to a sub-optimal attack.

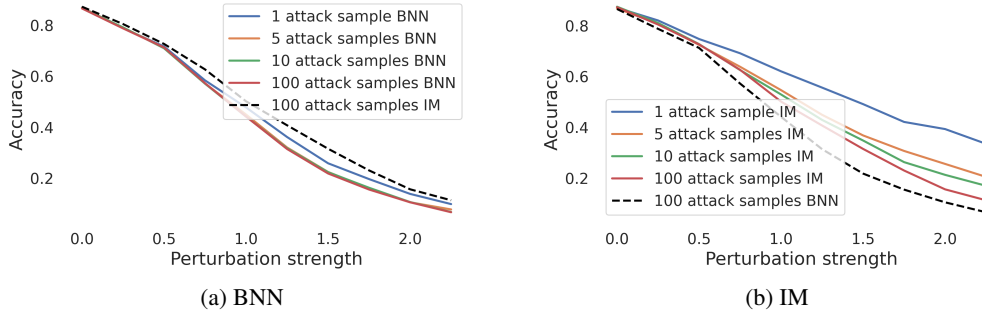


Figure 12: Accuracy under PGD attack with 100 iterations for a) the BNN and b) the IM on the first 1,000 test set images from FashionMNIST for different perturbation strengths and amount of samples used for calculating the attack.

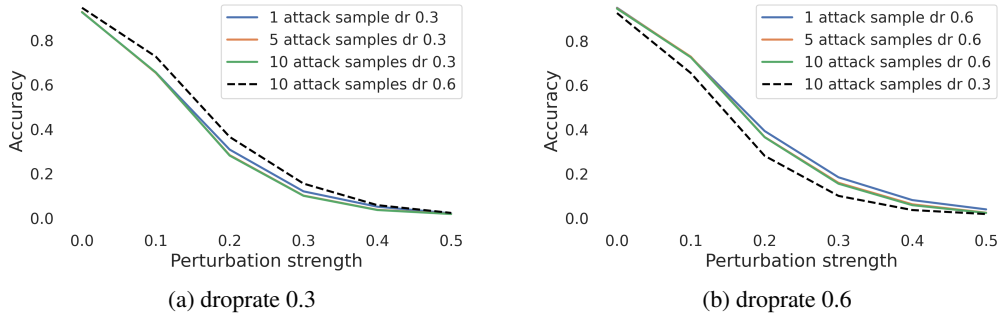


Figure 13: Accuracy under PGD attack with 100 iterations for the ResNet with a dropout probability of a) 0.3 and b) of 0.6 on CIFAR10 for different perturbation strengths and amount of samples used for calculating the attack.

C.3 Discussing the impact of extreme prediction values

Attacks based on softmax predictions can be unsuccessful for deterministic and stochastic neural networks alike when encountering overly confident predictions. That is, predictions where the softmax output is equal to 1, since these lead to zero gradients. A practical solution to circumvent this problem is to calculate the gradient based on the logits [Carlini and Wagner, 2017a]. This approach is feasible for classifier whose predictions do not depend on the scaling of the output, that is, outputs which are equally expressive in both intervals $[0, 1]$ and $[-\infty, \infty]$. In Bayesian neural networks, where $f_c(x, \Theta)$ is per definition a probability the shortcut over taking the gradient over logits leads to distorted gradients. For completeness, we nevertheless look at the performance of an attack based on the logits for the two different models trained on FashionMNIST with softmax outputs: IM and BNN. We conducted an adversarial FGM attack based on 100 samples, but instead of using the cross-entropy